

Composite functions



1

a $f(-2) = -32$

b $g(f(-2)) = -27$

c $f(g(0)) = 500$

2 $f(g(2)) = 8$

3

a $f(7) = -20$

b $g(f(7)) = -107$

c $g(7) = 28$

d $f(g(7)) = -62$

e No

4

a $f(g(x)) = (9x - 4)^3$

b $f(g(x)) = (x^2 + 1)^2$

c $f(g(x)) = \sqrt{4x - 3}$

d $f(g(x)) = 4x^3 + 9$

e $f(g(x)) = 4x^2 - 3$

5

a $g(6) = 136$

b $f(g(6)) = -270$

c $r(f(g(6))) = 802$

6

a $g(f(6)) = 1596$

b $h(x) = -8x^2$

c Quadratic

7

a $r(x) = -2x^2 + 6$

b $q(x) = -6x^2 + 19$

8

a $f(2x) = 4x^2 + 3$

b No solution provided.

9

a Quadratic function

b $g(x) = 5x^2$

c $g(f(x)) = 45x^2 - 600x^3 + 2000x^4$

10

a $r(x) = x^2 + 6x + 8$

b No solution provided.

c $p(q(x)) = x^2 + 2$

d $x < -1$

11

$$\text{a } f(x)g(x) = 4x + \frac{12}{x}$$

$$\text{c } \frac{f(x)}{g(x)} = \frac{4}{x(x^2 + 3)}$$



$$\text{b } f(g(x)) = \frac{4}{x^2 + 3}$$

$$\text{d } \frac{g(x)}{f(x)} = \frac{x(x^2 + 3)}{4}$$

12

$$\text{a } f(x)g(x) = (9x + 4)(x + 1)(x - 3)$$

$$\text{c } \frac{f(x)}{g(x)} = \frac{9x + 4}{x^2 - 2x - 3}$$

$$\text{b } f(g(x)) = 9x^2 - 18x - 23$$

$$\text{d } \frac{g(x)}{f(x)} = \frac{x^2 - 2x - 3}{9x + 4}$$

Domain and range

13

$$\text{a } \text{i } f(g(x)) = \frac{9}{1 - \frac{6}{x}}$$

$$\text{iii } g(f(x)) = \frac{1 - 2x}{3}$$

$$\text{ii } (-\infty, 0) \cup (0, 6) \cup (6, \infty)$$

$$\text{iv } \left(-\infty, \frac{1}{2}\right) \cup \left(\frac{1}{2}, \infty\right)$$

$$\text{b } \text{i } f(g(x)) = |x|$$

$$\text{iii } g(f(x)) = x$$

$$\text{ii } (-\infty, \infty)$$

$$\text{iv } [-4, \infty)$$

$$14 \quad f(x) = \sqrt{1 + x}$$

15

$$\text{a } f(x) = \frac{1}{x^6}$$

$$\text{b } g(x) = (x - 2)^6$$

$$16 \quad g(x) = x$$

17



a

i $x \in (-\infty, \infty), y \in [0, \infty)$

ii $x \in (-\infty, \infty), y \in (-\infty, \infty)$

iii $x \in (-\infty, \infty), y \in [0, \infty)$

b

i $x \in [0, \infty), y \in [0, \infty)$

ii $x \in (-\infty, \infty), y \in (-\infty, \infty)$

iii $x \in [3, \infty), y \in [0, \infty)$

c

i $x \in [0, \infty), y \in [0, \infty)$

ii $x \in (-\infty, \infty), y \in (-\infty, \infty)$

iii $x \in \left[-\frac{9}{4}, \infty\right), y \in [0, \infty)$

d

i $x \in (-\infty, \infty), y \in (-\infty, \infty)$

ii $x \in (-\infty, \infty), y \in (-\infty, \infty)$

iii $x \in (-\infty, \infty), y \in (-\infty, \infty)$

18

a $f(g(x)) = \sqrt{5x-1}$

b $x \geq \frac{1}{5}$

c $g(f(x)) = 5\sqrt{x+5} - 6$

d $x \geq -5$

19

a $f(g(x)) = -\frac{2x}{x-5}$

b $x = 5$

c $g(f(x)) = \frac{2x}{2x+5}$

d $x = -\frac{5}{2}$

Applications

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a $g(x) = \frac{3}{4}x$ or $g(x) = 0.75x$

b $f(x) = x - 5$

c $f(g(x)) = \frac{3}{4}x - 5$ or $f(g(x)) = 0.75x - 5$

d \$65.50

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a $A(m) = 1.6m + 17.6$

b \$53.60